Part 4. Architecting on AWS

Werner Vogels, CTO of Amazon.com, is quoted as saying "Everything fails all the time." Instead of trying to reach the unreachable goal of an unbreakable system, AWS plans for failure in the following ways:

- Hard drives can fail, so S3 stores data on multiple hard drives to prevent the loss of data.
- Computing hardware can fail, so virtual machines can be automatically restarted on another machine if necessary.
- Data centers can fail, so there are multiple data centers per region that can be used in parallel or on demand.

Outages of IT infrastructure and applications can cause a loss of trust and money and are a major risk for businesses. You will learn how to prevent an outage of your AWS applications by using the right tools and architecture.

Some AWS services handle failure by default in the background. For some services, responding to failure scenarios is available on demand. And some services don't handle failure by themselves but offer the possibility to plan and react to failure. The following table shows an overview of the most important services and their failure handling.

Designing for failure is a fundamental principle of AWS. Another is to make use of the elasticity of the cloud. You will learn about how to